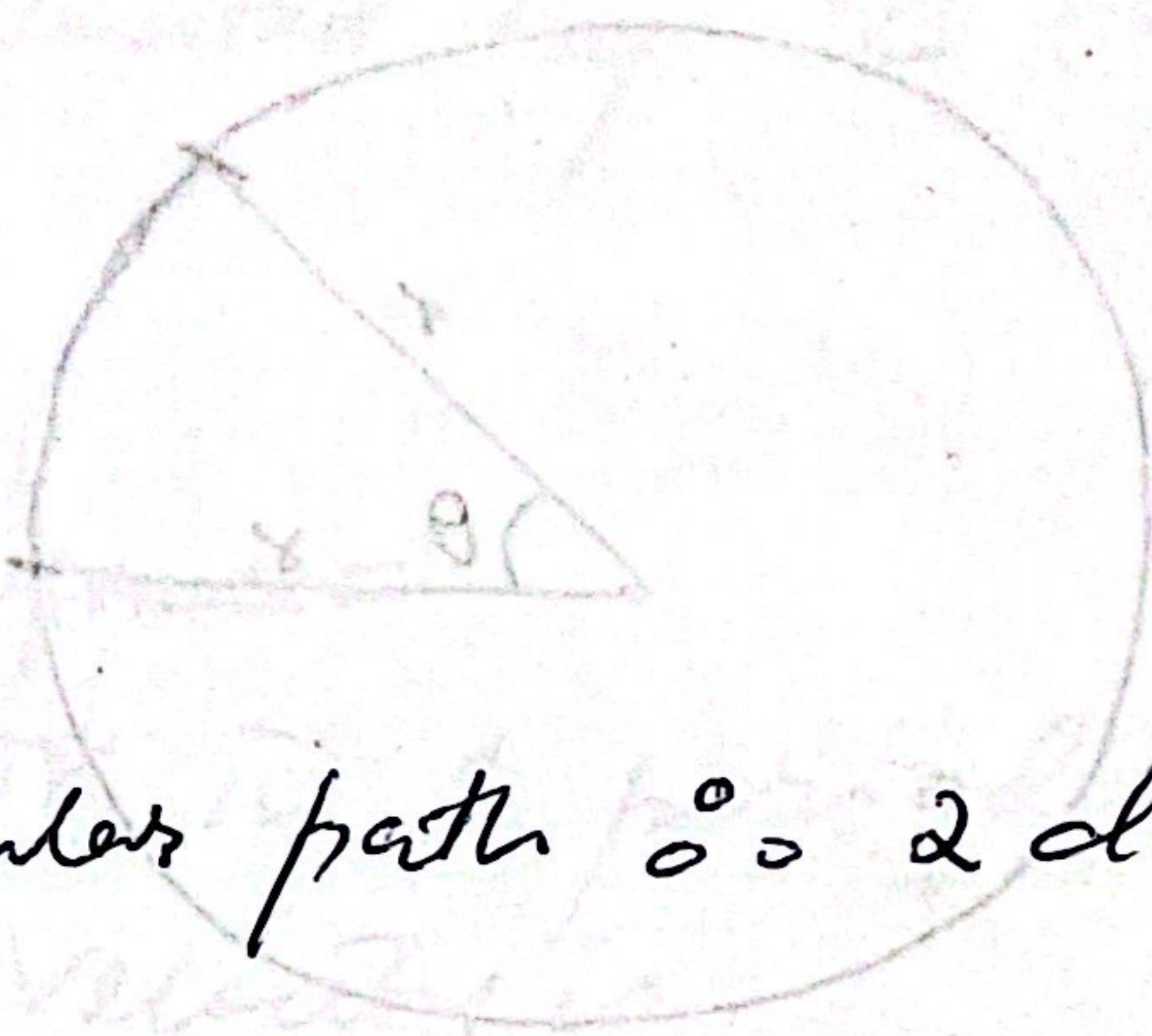
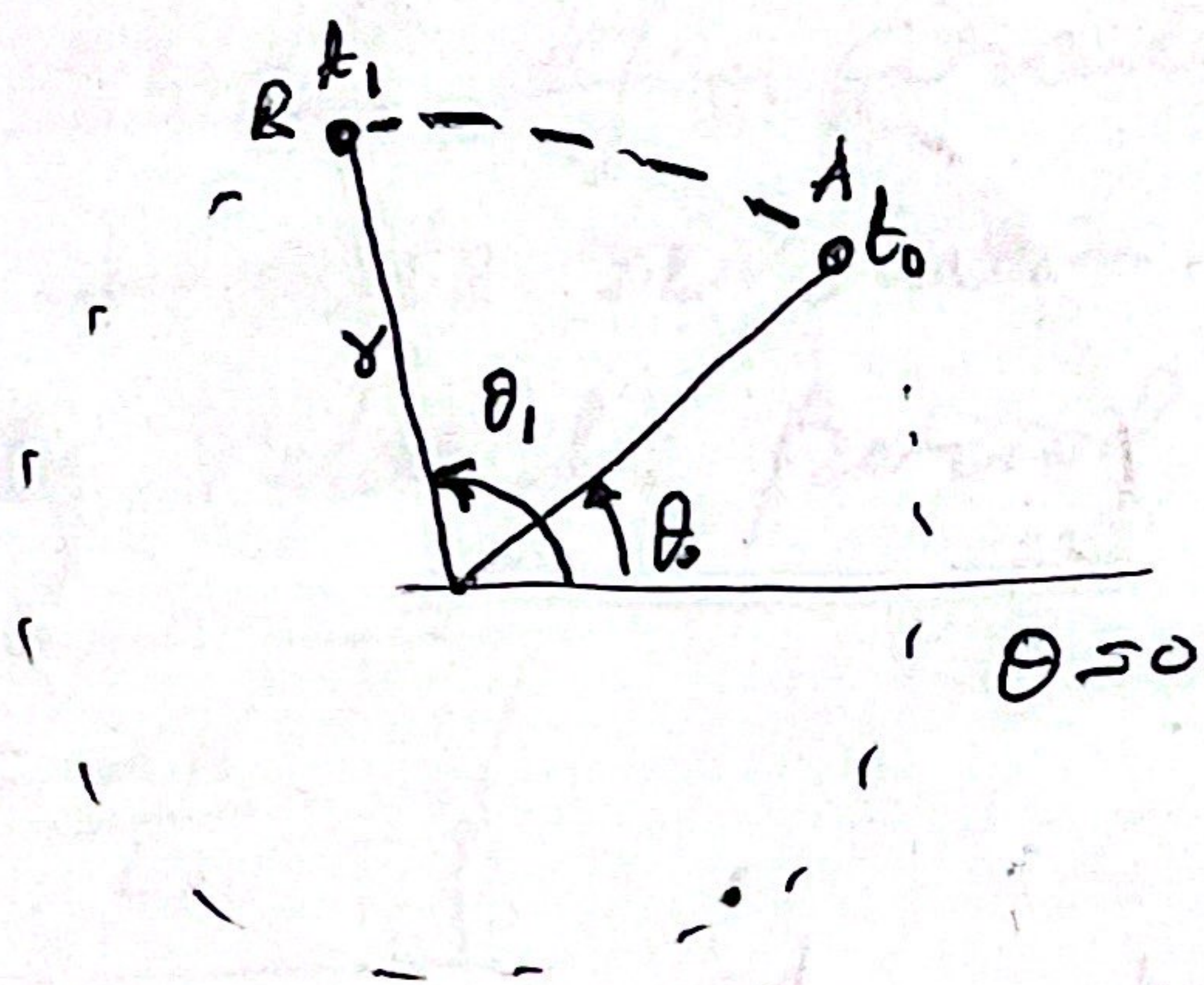


Circular motion — 2D motion in a 2D plane.



This is a motion along a circular path i.e. 2 dimensional.
in a coordinate system.

Assumptions

□ uniform motion. i.e. speed constant.

1 complete revolution = 2π radians

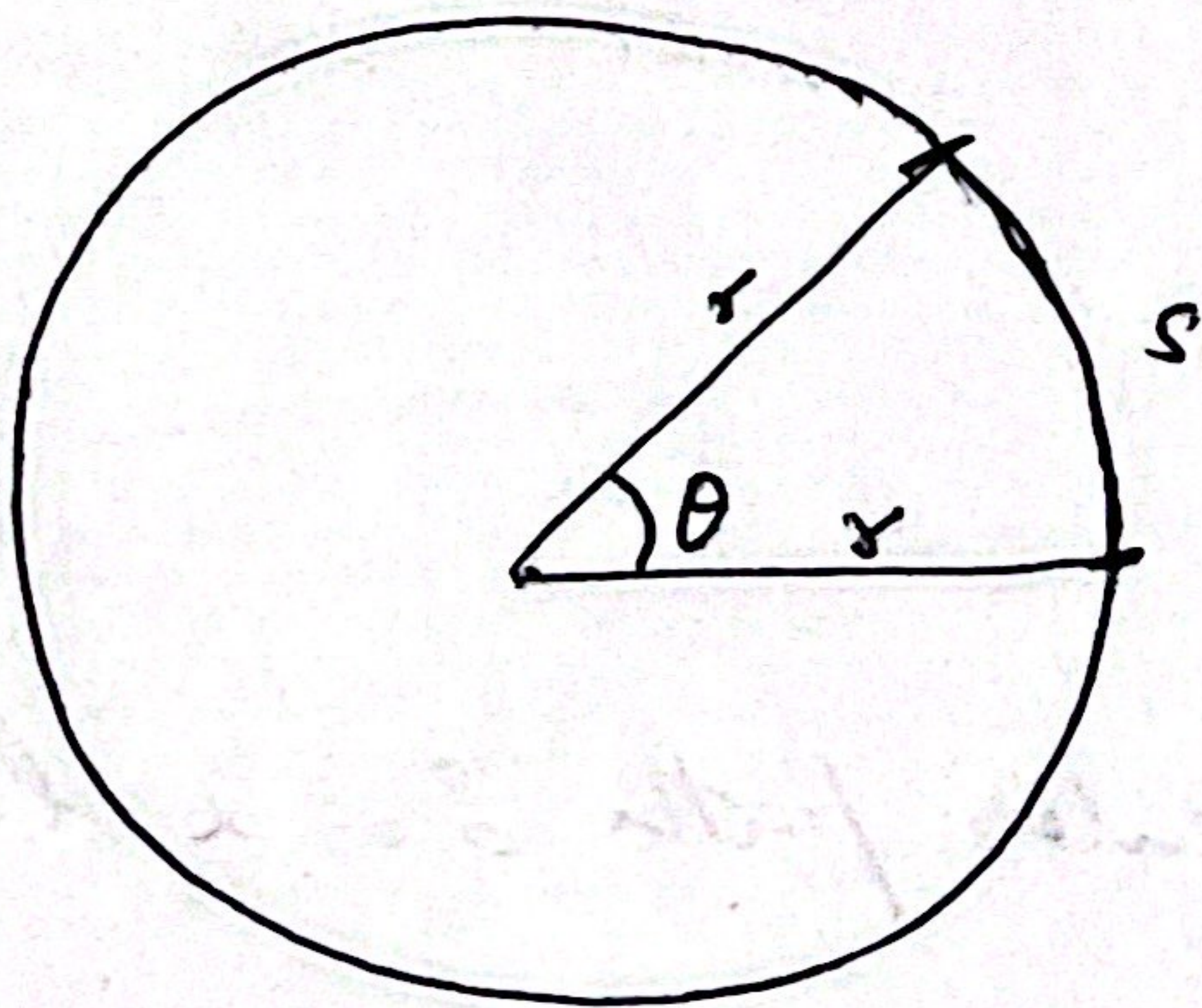
If an object moves from point A identified by θ_0 to point B identified by θ_1 in a time interval $t_1 - t_0$, the average angular velocity is

$$\bar{\omega} = \frac{\theta_1 - \theta_0}{t_1 - t_0} = \frac{\Delta\theta}{\Delta t} \text{ rad/s.}$$

Analogous to the ^{linear} instantaneous velocity, the instantaneous angular velocity is defined to be

$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t}$$

Note



$s = r\theta$ θ measured in radians.

$$\text{Circumference} = 2\pi r$$

Therefore 360° in a complete circle $\cong 2\pi$ radians

Thus

$$1 \text{ radian} = \frac{360^\circ}{2\pi} \cong 57.3^\circ$$

$$v = \frac{d}{t}$$

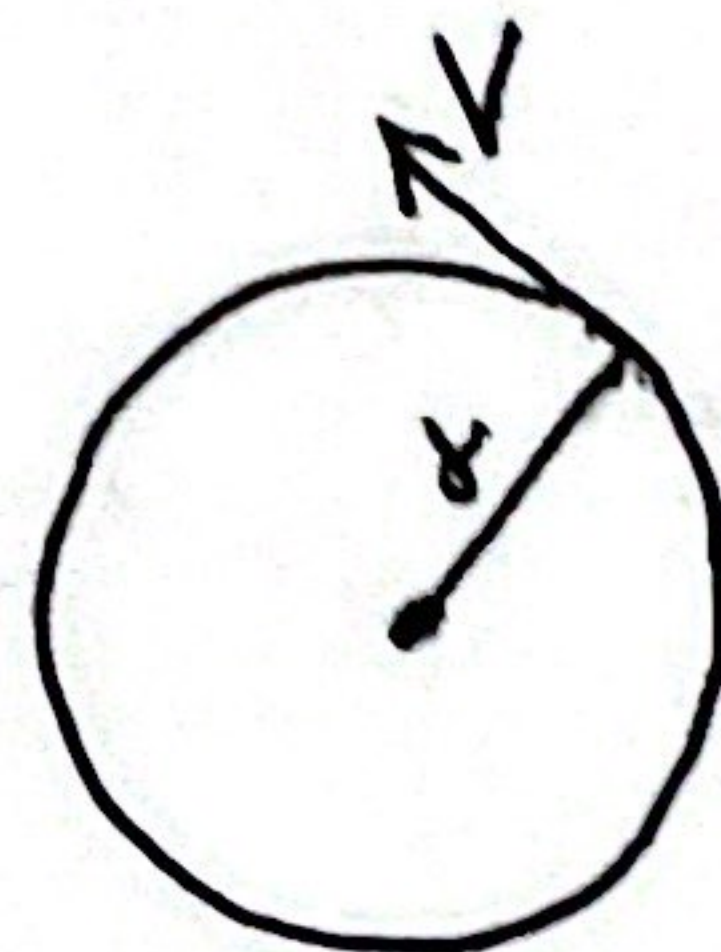
$$t = \frac{d}{v}$$

The period of a circular motion is the time required for 1 complete revolution. This period and ω are inversely related. Period is denoted by τ (tau) and is given by

$$\tau = \frac{2\pi}{\omega}$$

The distance covered is ^{the circumference =} $2\pi r$ in 1 period or τ . Therefore the magnitude of the linear velocity is

$$V = \frac{\text{distance}}{\text{time}} = \frac{2\pi r}{2\pi/\omega}$$



or
$$V = r\omega$$

linear velocity V = product of radius and angular velocity ω

Example

An automobile moves with a constant speed of 50 km/hr around a track of 1 km in diameter. What is ω and τ of the motion?

Sol.

$$\omega = \frac{V}{r} = \frac{50 \text{ km/hr}}{0.5 \text{ km}} = 100 \text{ rad/hr}$$

$$\tau = \frac{2\pi}{\omega} = \frac{2\pi}{100 \text{ rad/hr}} = 0.063 \text{ hr} = 3.8 \text{ min.}$$

Angular acceleration.

Angular acceleration $\boxed{a_c = \frac{v^2}{r}}$ $r = \frac{v^2}{a}$

The direction of acceleration is directed toward the Centre of rotation. The acceleration is "Centre seeking" and is termed Centripetal acceleration.

Note Since $v = r\omega$

therefore $a_c = \frac{v^2}{r} = \frac{(r\omega)^2}{r} = r\omega^2$

$$\boxed{a_c = r\omega^2}$$

[Alternative expression for a_c]

Example

What are v , ω and a_c for the motion of the Earth around the Sun? $r = 1 \text{ A.U.} = 1.5 \times 10^{11} \text{ m}$; $T = 1 \text{ yr} = 3.15 \times 10^7 \text{ s}$

Sol.

$$\omega = \frac{2\pi}{T} = \frac{2\pi \text{ rad}}{3.15 \times 10^7 \text{ s}} \approx 2 \times 10^{-7} \text{ rad/s}$$

$$v = r\omega = 1.5 \times 10^{11} \text{ m} \times 2 \times 10^{-7} \text{ rad/s}$$

$$= 3 \times 10^4 \text{ m/s}$$

$$= 30 \text{ km/s}$$

$$a_c = r\omega^2 = 1.5 \times 10^{11} \text{ m} \times (2 \times 10^{-7} \text{ rad/s})^2$$

$$= 0.006 \text{ m/s}^2$$

Note The acceleration towards the centre of the sun is

$$\frac{0.006 \text{ m/s}^2}{9.8 \text{ m/s}^2} \approx 0.06\%$$

7*

Exercise

Q2 Tu12

□ An object is moving in a circular path of radius 9.4m experiences an acceleration of $3g$. What is the speed of the object in m/s ? What is its period of function.

□

Important

There is a similarity between linear and circular motions. That is, every relation for a linear motion corresponds to one for circular motion.

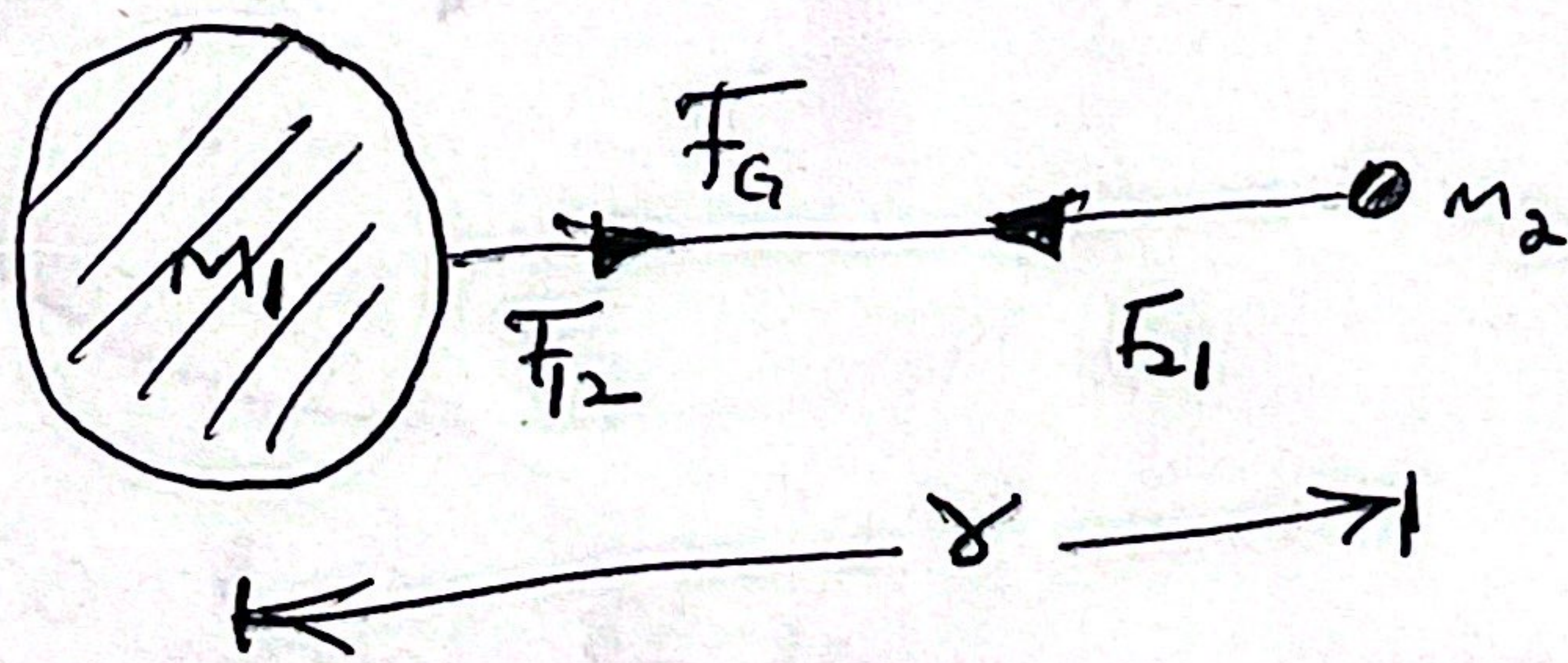
	Linear	Circular
Displacement	x	θ
Speed	$v = \frac{\Delta x}{\Delta t}$	$\omega = \frac{\Delta \theta}{\Delta t}$
Acceleration	$a = \frac{\Delta v}{\Delta t}$	$\alpha = \frac{\Delta \omega}{\Delta t}$
	$v = v_0 + at$	$\omega = \omega_0 + \alpha t$
	$x = v_0 t + \frac{1}{2} at^2$	$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$
	$v^2 - v_0^2 = 2ax$	$\omega^2 - \omega_0^2 = 2\alpha\theta$

The universal Gravitational force law.

states: The gravitational force which two particles or spherical objects mutually exert on one another is inversely proportional to the square of the distance between their centres and directly proportional to the product of their masses.

$$F_g = G \frac{M_1 M_2}{r^2}$$

G is a constant of proportionality $G = 6.673 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$ and is called the universal gravitational constant.



This force is responsible for the motion of planets around ~~the star~~ and satellites around the planets. In general, the motion of stars in galaxies is governed by this law.

Example.

With what force does the Earth and the moon attract

$$r_m = 3.84 \times 10^8 \text{ m}$$

$$M_m = 7.35 \times 10^{22} \text{ kg}$$

$$M_E = 5.98 \times 10^{24} \text{ kg}$$

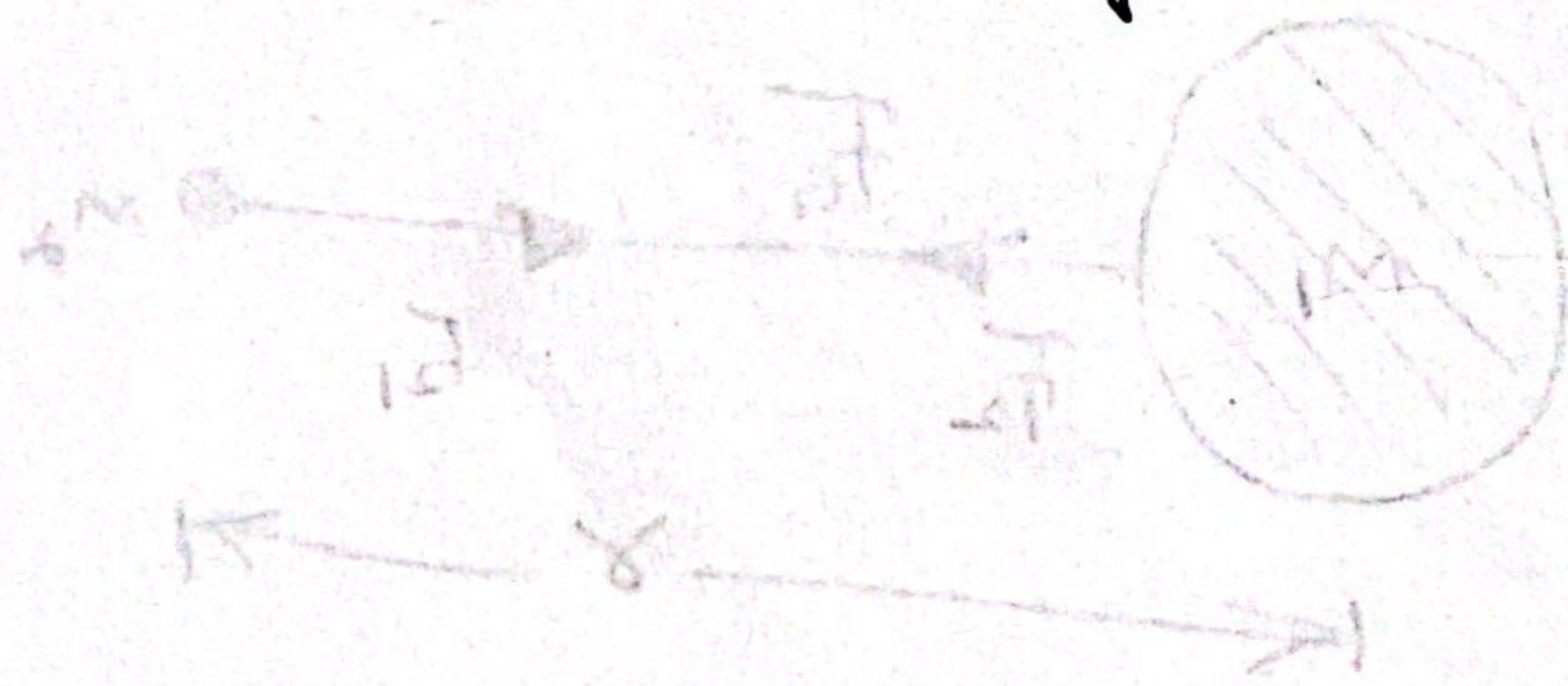
Sol.

$$F_G = \frac{G M_m M_E}{r_m^2} = (6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2) \times \frac{7.35 \times 10^{22} \text{ kg} \times 5.98 \times 10^{24} \text{ kg}}{(3.84 \times 10^8)^2}$$
$$= 2.0 \times 10^{20} \text{ N.}$$

We can obtain this result in another way. If we know the moon's centripetal acceleration then

$$F_G = M_m a_c = M_m \frac{v^2}{r_m} = 7.35 \times 10^{22} \text{ kg} \times 0.009 \text{ m/s}^2 \times 0.30$$

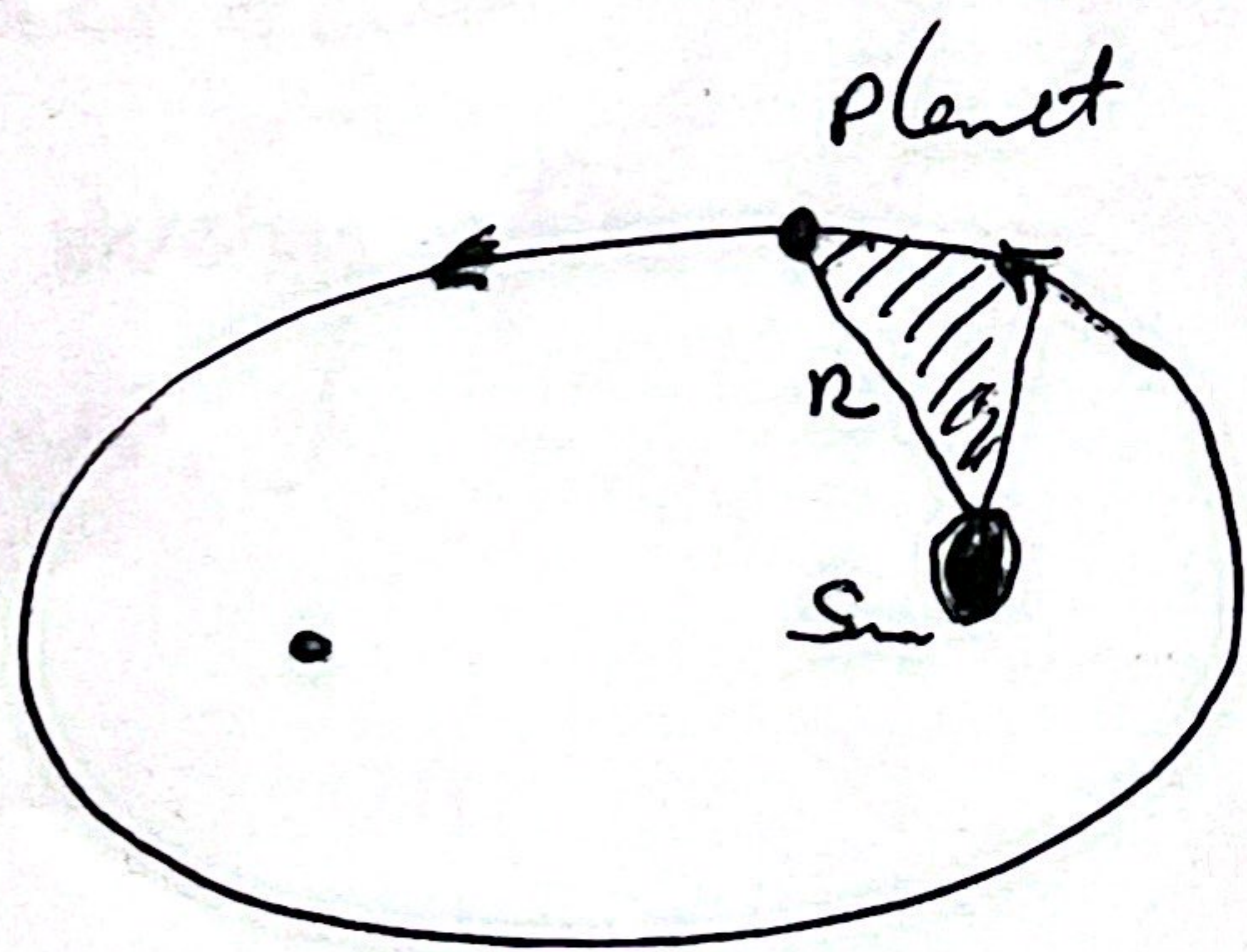
$$\approx 2 \times 10^{20} \text{ N}$$



Planetary motion

Kepler's laws

- I. The motion of a planet is an ellipse with the Sun at one focus.
- II. The line connecting the planet with the Sun sweeps out equal areas in equal times.
- III. The period of a planet's motion and its distance from the Sun are related by $\frac{R^3}{T^2} = \text{constant}$
Where the constant is the same for all planets.
[The square of the period is $\propto$ to the cube of the length of semi major].



The second law of Kepler expresses the fact that the angular momentum of the Earth around the Sun remains constant.

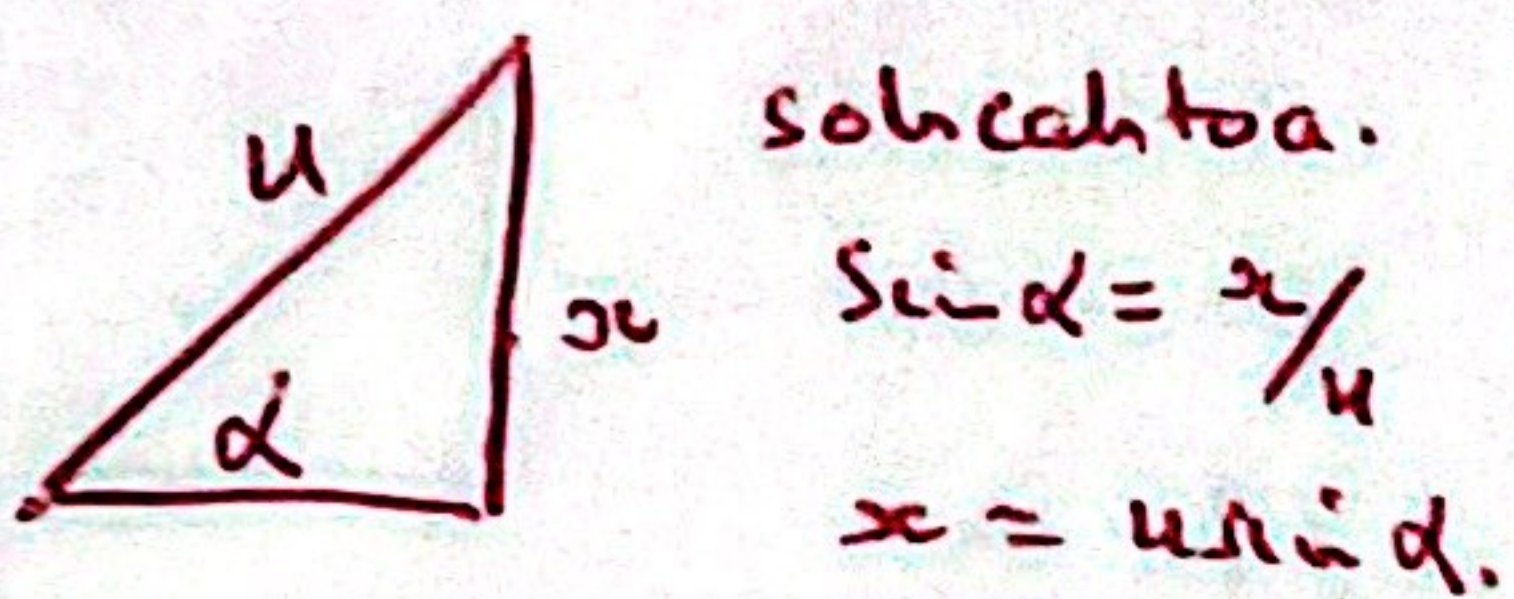
Even though the first law describes the Sun as at one focus of an ellipse, it is not the centre of the Sun that lies at one focus but rather the centre of mass of the sun-planet system. And because the Sun is much more massive

than any planet, the centre of mass of any sun-planet combination lies very near the centre of the Sun.

Although a small relativistic correction in Newton's form of gravitational force law, is required for objects in motion with high velocities, the Newtonian equation is valid for essentially all practical purposes and it appears to be entirely correct for static gravitational forces.

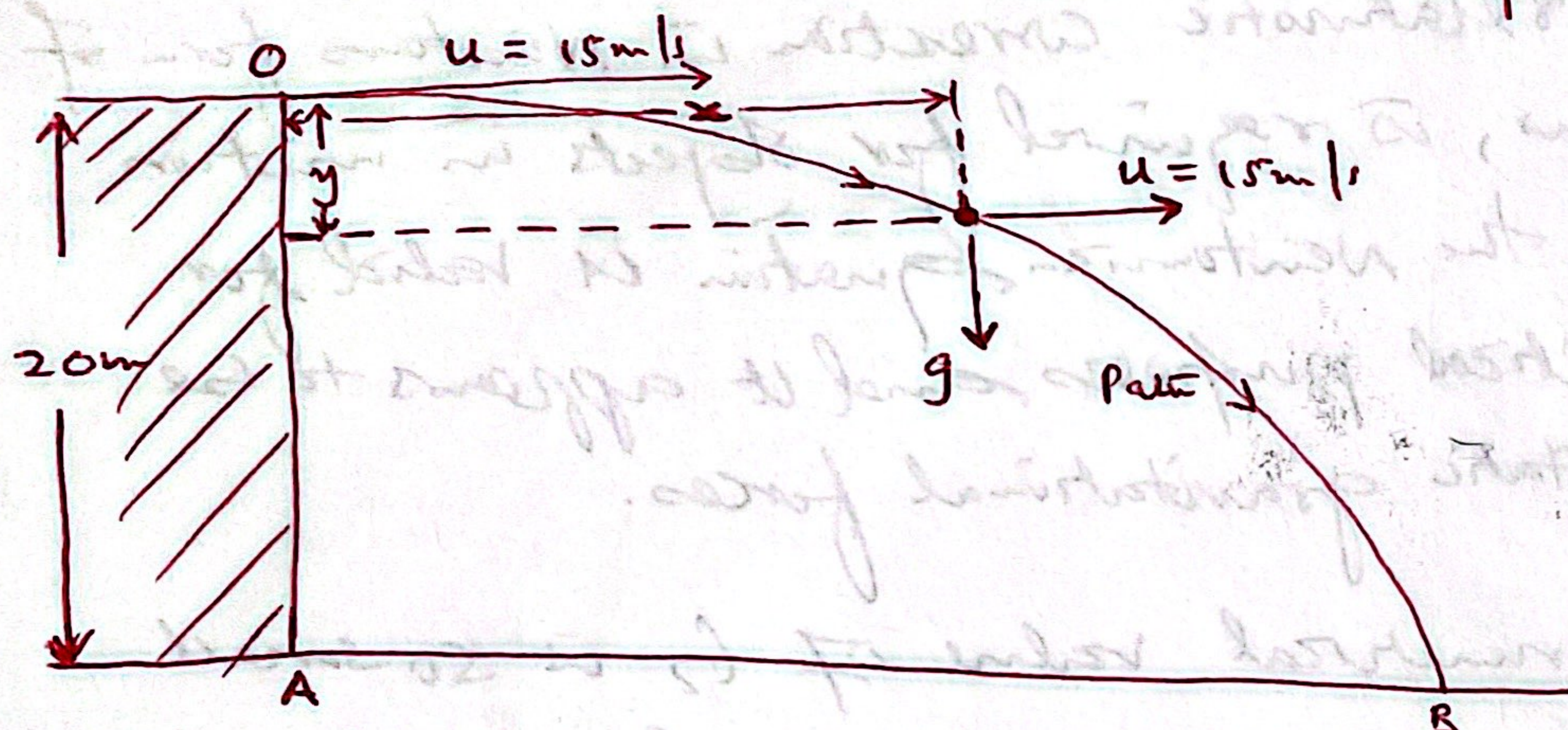
In SI units, the numerical value of G is so small that two equal masses of 1 kg separated 1 m attract each other with a force of 6.7×10^{-11} N. Too negligible.

In the equation, the masses in the numerator are the measures of ~~the~~ the strength of attraction. The Sun itself has a mass that greatly exceeds that of any planet and therefore, every planet responds to the sun and ignores its fellow planets.



Projectile

- Consider an object thrown forward from the top of a cliff OA with a horizontal velocity u of 15 m/s .



- Since u is horizontal, it has no component in a vertical direction. Similarly, since g acts vertically, it has no component in a horizontal direction.
- (i) Vertical motion. If $OA = 20 \text{ m}$ the ball has no vertical velocity. Vertical acc = $g = 9.8 \text{ m/s}^2$.

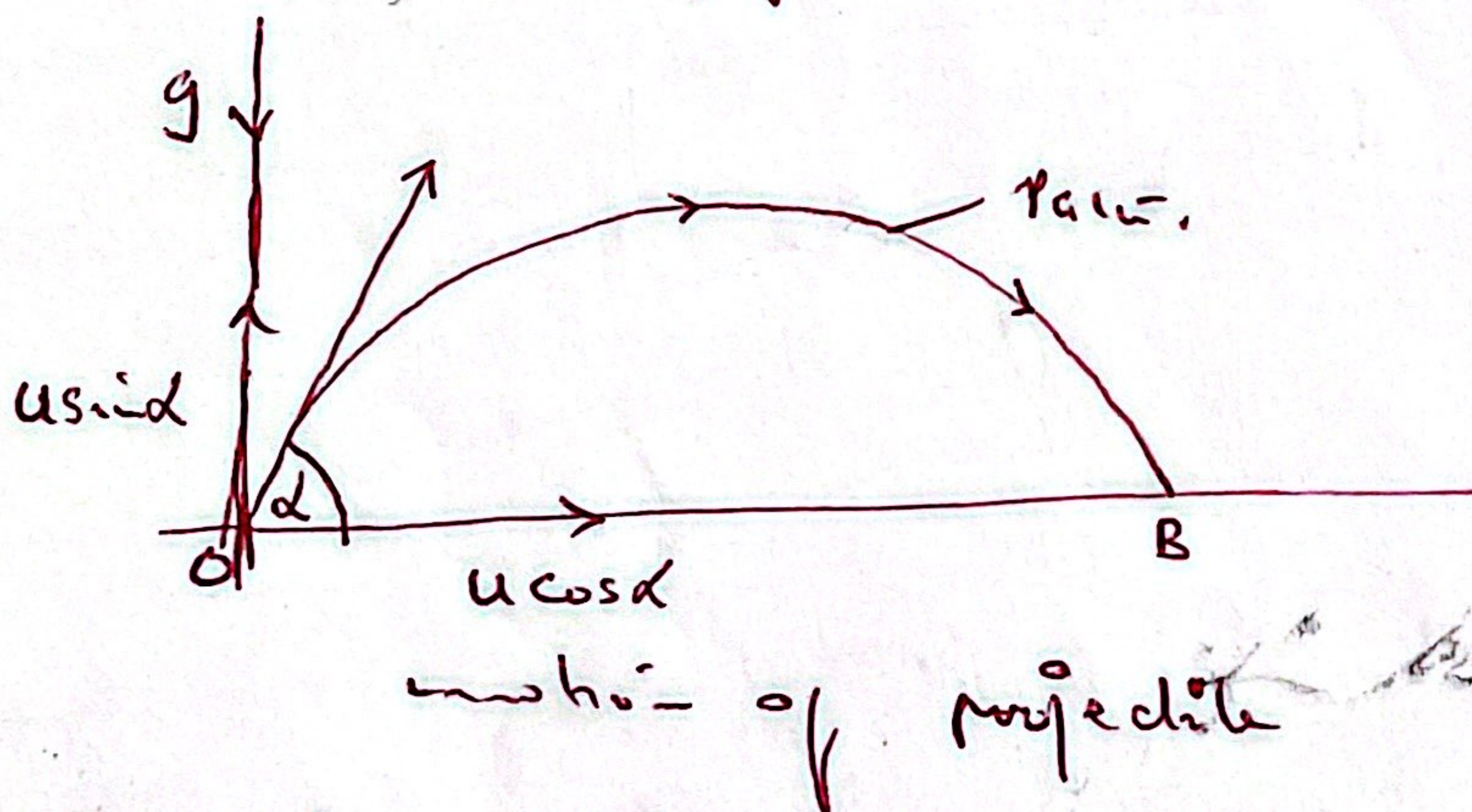
$$s = ut + \frac{1}{2}gt^2 \quad 20 = \frac{1}{2} \times 10 \times t^2 = 5t^2 \quad t = 2 \text{ s.}$$

- (ii) horizontal distance - Ball moves in constant horizontal velocity of 15 m/s .

$$\text{horizontal distance AB} = \text{distance from cliff} = 15 \times 2 = 30 \text{ m.}$$

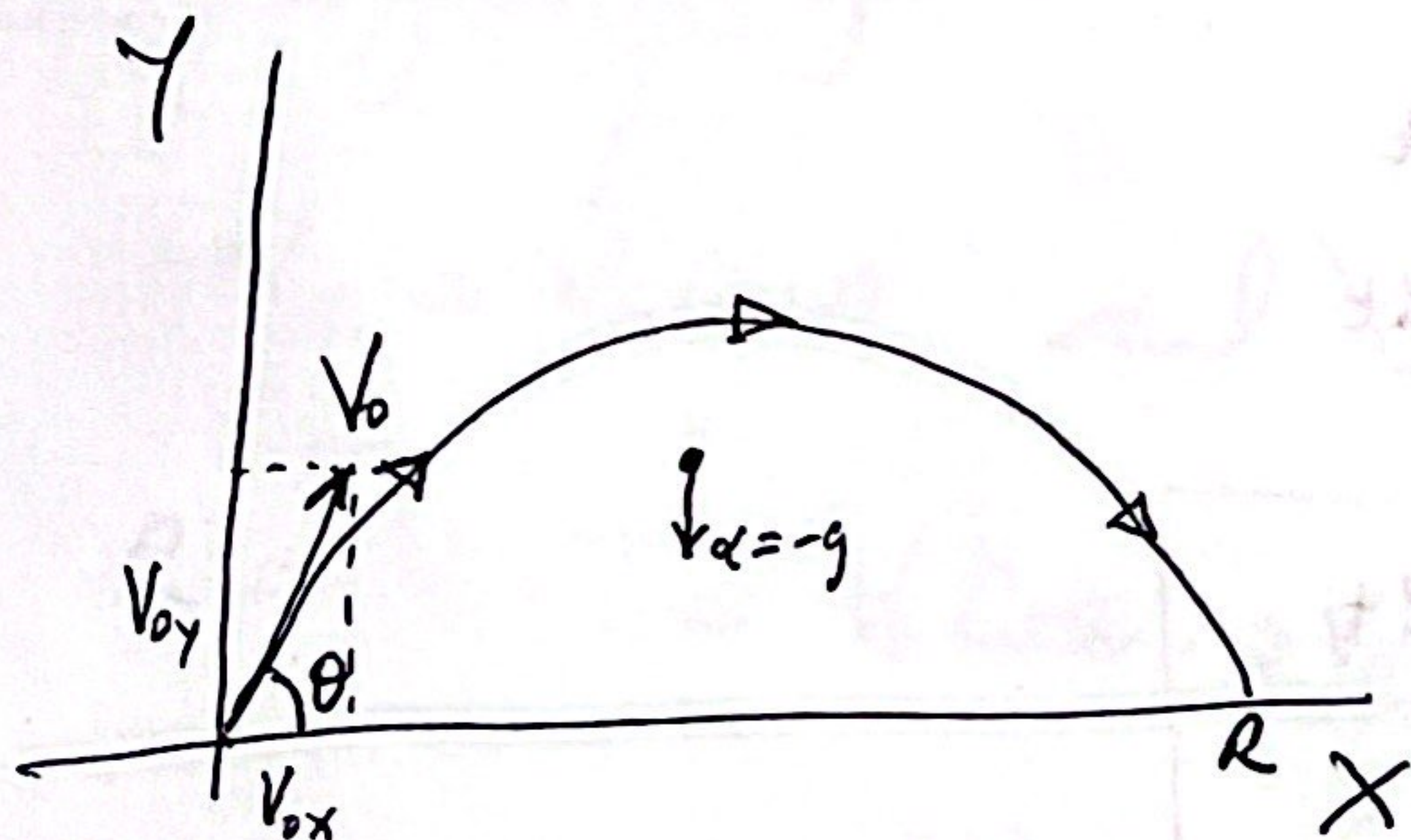
Motion of Projectiles and their Range

- consider the fig below.



- (i) Vertical motion: The vertical component of u is $u \sin \alpha$; the acc $a = -g$. When the projectile reaches the ground at B, the vertical distance s

projectile motion.



$$V_{0x} = V_0 \cos \theta$$

$$V_{0y} = V_0 \sin \theta$$

Using the equations of linear motion namely

$$V = V_0 + at$$

$$X = V_0 t + \frac{1}{2} at^2$$

then

$$V_x = V_{0x}$$

$$= V_0 \cos \theta$$

and

$$V_y = V_{0y} + at$$

$$= V_{0y} - gt$$

$$V_y = V_0 \sin \theta - gt$$

Time of flight

— The projectile goes up and falls back. It takes half time to go up to maximum height V_y and another half to go down V_y . Total $V_y = 0$

$$0 = V_0 \sin \theta - gt$$

$$\Rightarrow t = \frac{V_0 \sin \theta}{g}$$

$$T = \frac{2V_0 \sin \theta}{g}$$

travelled is zero. So, from $S = ut + \frac{1}{2}gt^2$ we have

$$0 = u \sin \alpha \cdot t - \frac{1}{2}gt^2$$

$$\frac{1}{2}gt^2 = u \sin \alpha \cdot t$$

$$gt = 2u \sin \alpha$$

$$t = \frac{2u \sin \alpha}{g}$$

$$T = \frac{2u \sin \alpha}{g}$$

(ii) Horizontal motion: Since g acts vertically, the horizontal component in a horizontal direction.

- So ball moves in a horizontal dir. $\approx$ an unchanged constant velocity $u \cos \alpha$. So

Range $R = OB = \text{Velocity} \times \text{time}$.

$$= u \cos \alpha \cdot t$$

$$= u \cos \alpha \times t \quad \text{but } t = \frac{2u \sin \alpha}{g}$$

$$= u \cos \alpha \times \frac{2u \sin \alpha}{g} = \frac{2u^2 \sin \alpha \cos \alpha}{g}$$

$$\text{but } 2 \sin \alpha \cos \alpha = \sin 2\alpha$$

$$R = \frac{u^2 \sin 2\alpha}{g}$$

max. Range: is obtained when $\sin 2\alpha = 1$, or

$2\alpha = 90^\circ$. So $\alpha = 45^\circ$. For max range E

a given velocity g into u . In this case

max range is $\frac{u^2}{g}$

- At max height A vertical $v = 0$. So applying $v = u + gt$ in a vertical dir. time to reach A is given by

$$0 = u \sin \alpha - gt \quad \text{or} \quad t = \frac{u \sin \alpha}{g} \quad \text{which is } \frac{1}{2} \text{ time to reach B.}$$

Range of a projectile — maximum horizontal distance

$$V_x = V_0 \cos \theta \quad \text{and} \quad \text{Range } R = V_x \cdot \text{total time } T.$$

$$R = V_x \cdot T = V_0 \cos \theta \cdot \frac{2V_0 \sin \theta}{g}$$

$$= \frac{V_0^2}{g} 2 \sin \theta \cos \theta$$

$$= \frac{V_0^2}{g} \sin 2\theta$$

$$[\sin 2\theta = 2 \sin \theta \cos \theta]$$

$$\boxed{R = \frac{V_0^2}{g} \sin 2\theta}$$

Note

When $\sin 2\theta = 1$ $R_{\max} = \frac{V_0^2}{g}$
 $\therefore \theta = 45^\circ$

Maximum height —

$$H = V_{y0} t = V_0 \sin \theta \cdot \frac{V_0 \sin \theta}{g}$$

$$= \frac{V_0^2}{g} \sin^2 \theta$$

$$\boxed{H = \frac{V_0^2}{g} \sin^2 \theta}$$

$$H = V_{y0} t - \frac{1}{2} g t^2$$

$$= V_0 \sin \theta \cdot \frac{V_0 \sin \theta}{g} - \frac{g \cdot V_0^2 \sin^2 \theta}{2g^2}$$

$$= \frac{V_0^2 \sin^2 \theta}{g} (1 - \frac{1}{2}) = \frac{V_0^2 \sin^2 \theta}{2g}$$

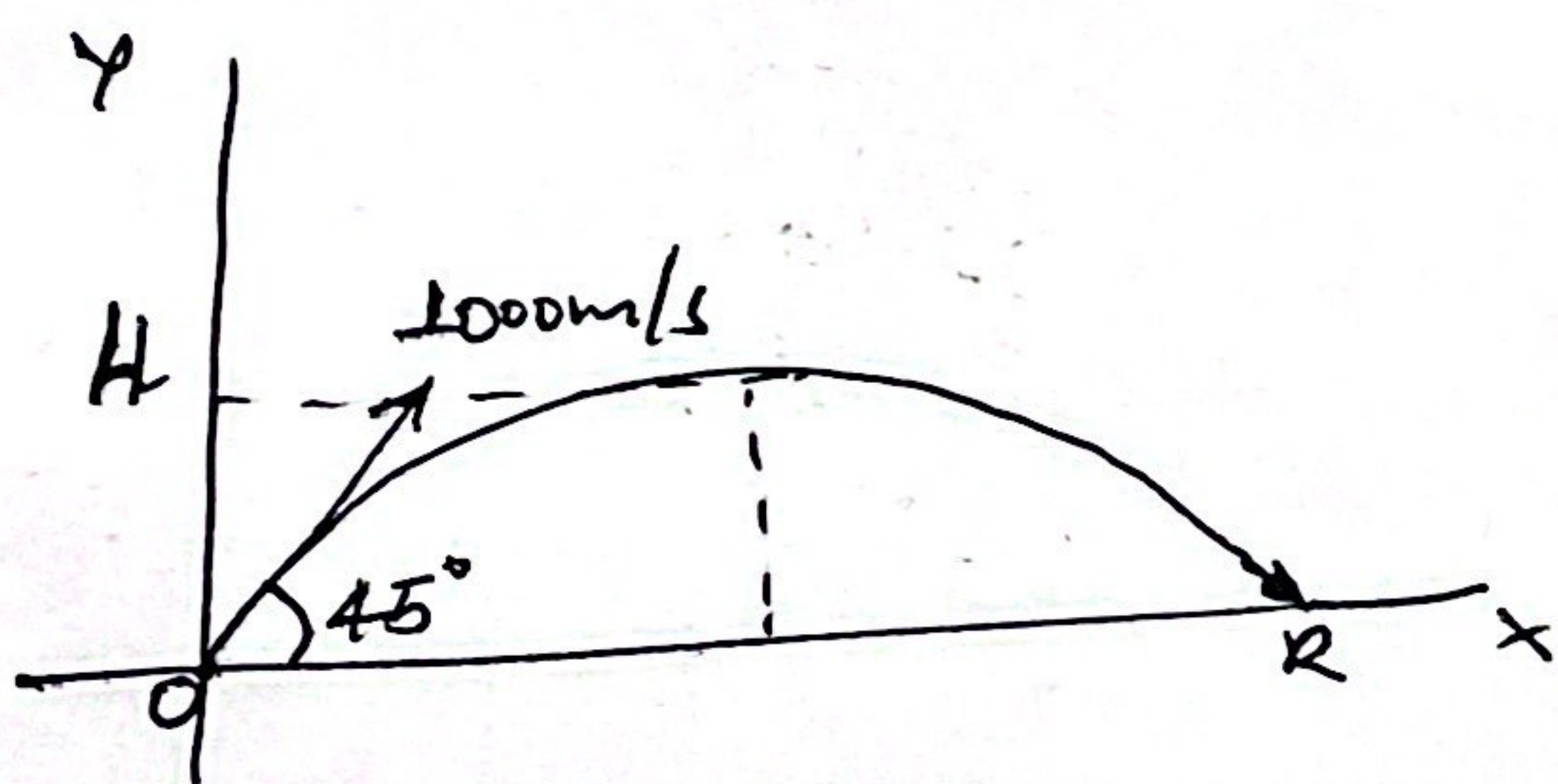
$$\boxed{H = \frac{V_0^2 \sin^2 \theta}{2g}}$$

Continue for max
height

Example.

A rifle is pointed at an angle of 45° above the horizontal and fires a bullet with a muzzle velocity of 1000 m/s . To what height does the bullet rise and what is its horizontal range? Estimate the time of flight.

Sol.



$$T = \frac{2V_0 \sin \theta}{g} = \frac{2 \times 1000 \text{ m/s} \times \sin 45^\circ}{10(9.8) \text{ m/s}^2} = 141.42$$

$$H = \frac{V_0^2 \sin^2 \theta}{2g} = \frac{(1000 \text{ m/s})^2 \times (\sin 45^\circ)^2}{10(9.8) \text{ m/s}^2} = \frac{10^6 \text{ m}^2/\text{s}^2 \times 1/2}{10(9.8) \text{ m/s}^2} \approx 2.5 \times 10^4 \text{ m}$$

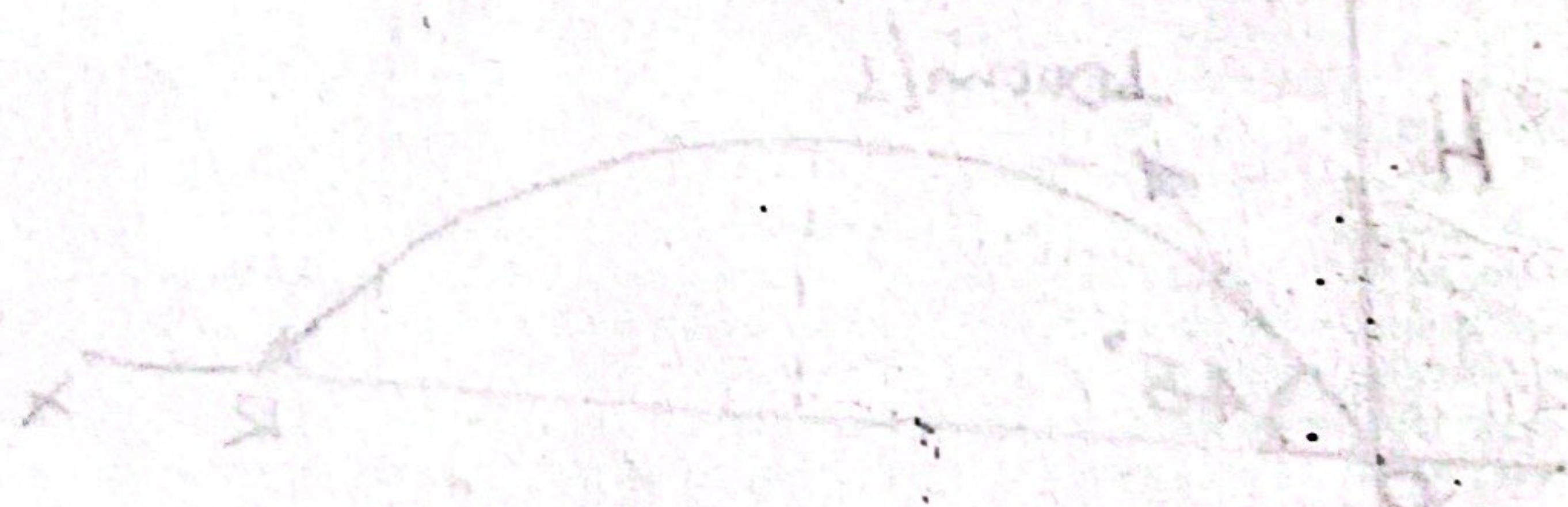
$$R = \frac{V_0^2 \sin 2\theta}{g} = \frac{(1000 \text{ m/s})^2 \times \sin 90^\circ}{10(9.8) \text{ m/s}^2} = \frac{10^6 \text{ m}^2/\text{s}^2 \times 1}{10(9.8) \text{ m/s}^2} = 10^5 \text{ m} = 10 \text{ km}$$

8 *

Exercise.

If there were no air friction, what would be the maximum range of a projectile fired at mach 1 (the speed of sound)?
At what angle should it be fired to achieve this range?

Hint These two questions are best answered in the reverse order.



$$T = \frac{2 \times 1000 \text{ ft}}{32 \text{ ft/s}^2} = 62.5 \text{ s}$$

$$R = \frac{(2 \times 1000 \text{ ft}) \times \sin(2\theta)}{32 \text{ ft/s}^2} = 1000 \text{ ft}$$

$$\sin(2\theta) = \frac{1000 \text{ ft} \times 32 \text{ ft/s}^2}{2 \times 1000 \text{ ft}} = 16$$